

Quantum Computing

Part 1 - Hardware by Modality: Superconducting Qubits

Compiled by the Spaced Research Ledger

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Superconducting qubits are tiny circuits, cooled to near absolute zero, that behave like artificial atoms. They are the most widely deployed way to build a quantum computer, and the approach behind most of the machines the public has heard of. They are fast, they are manufacturable using methods borrowed from the chip industry, and they lose their information quickly. Everything in this part follows from that last trade.

Two measurements recur throughout. Coherence time is how long a qubit holds its state before noise destroys it, usually written as T1 and T2 and measured in microseconds. Gate fidelity is how often an operation does what it was supposed to, quoted as a percentage that matters mainly in how many nines follow the decimal point.

Section 1 covers IBM, Section 2 Google, Section 3 Rigetti, Section 4 IQM, and Section 5 the Fujitsu and RIKEN collaboration. Section 6 covers materials research, where the next generation of these devices is being decided.

1. IBM

IBM has stopped competing on qubit count. Its current design deliberately uses fewer qubits than its predecessor and spends the difference on connectivity. The theory is that circuit depth, not qubit population, is what limits useful work.

Recent Developments

The first processor built on that design reached customers in January 2026. IBM's `ibm_miami` became available on the company's Premium and Flex cloud plans, making the Nighthawk design generally accessible rather than a laboratory demonstration [1].

Current State

Nighthawk was unveiled in November 2025 as IBM's first chip with a square qubit topology [2]. It raised the number of tunable couplers from 176 on the earlier Heron design to 218, which improves connections between neighbouring qubits and cuts the need for SWAP gates. A SWAP gate moves information between qubits that are not directly connected, and every one of them is overhead rather than useful work.

The chip carries 120 physical qubits [3]. That is fewer than Heron r2 before it, and the reduction is deliberate. The design trades qubit count for circuit-depth capacity.

Nighthawk-based systems support circuits requiring up to 5,000 two-qubit gates at launch [4]. Two-qubit gates are the entangling operations that do the actual computational work, so this figure is the practical measure of what the machine can run. IBM states it is currently the only company able to deliver accurate

results at that circuit complexity [5]. It attributes the capability to hardware combined with error-mitigation techniques, rather than to device fidelity alone.

2. Google Quantum AI

Google builds fewer machines than IBM and publishes more about each one. Its Willow chip is the device behind the field's below-threshold error-correction result, and its specifications are the most thoroughly documented in this part.

Current State

Willow was announced in December 2024 and carries 105 superconducting transmon qubits [6]. Google fabricates it end to end at its own Santa Barbara facility rather than using an outside foundry, which is unusual in an industry that mostly relies on contract manufacturing.

The chip's headline achievement is coherence. Willow's T1 relaxation time approaches 100 microseconds, roughly five times better than Google's previous Sycamore chip, which the company attributes to materials and fabrication changes that lower loss [6]. Google reports single-qubit gate fidelity near 99.97% and two-qubit gate fidelity near 99.88% [7].

Detailed characterization gives a more textured picture than the headline numbers. Average T1 across the 105-qubit processor runs from about 37 to 86 microseconds, and T2 from about 17 to 78 microseconds [9]. Two-qubit CZ gates execute in roughly 42 nanoseconds. A 72-qubit Willow variant does better on both counts, with T1 from about 75 to 85 microseconds and CZ gates in 37 nanoseconds. Smaller chips are easier to make uniform, and the spread across a larger device is itself a scaling problem.

Two architectural choices support those numbers. Google activates tunable couplers between adjacent qubits to perform two-qubit gates, replacing the passive capacitive coupling used in earlier chips such as Bristlecone, which reduced crosstalk and improved fidelity [8]. It also uses an inverted flip-chip integration style, bonding a separate wiring chip to the qubit chip [7]. That frees the qubit layer from dense on-chip routing, which is one of the practical limits on how many qubits fit in one device.

3. Rigetti

Rigetti's distinguishing bet is speed and modularity. Its gates are roughly a thousand times faster than the modalities in Parts 2 and 3. It builds large processors by connecting small chips rather than fabricating one big one.

Recent Developments

Rigetti's second-quarter 2026 report gave precise figures for its 108-qubit Cepheus-1-108Q system [10]. Median single-qubit gate fidelity is approximately 99.9%, median two-qubit gate fidelity approximately 99.1%, and gate speeds approximately 60 nanoseconds.

Current State

An independent result matters more than a vendor's own calibration. Quantum Machines operated a Rigetti Novera processor using its own control hardware and calibration software [11]. It reproduced a median two-qubit gate fidelity of 99.5% across 11 qubit pairs, peaking at 99.86%. It also measured 99.93% median single-qubit fidelity and 95% readout fidelity. The figures matched Rigetti's in-house results using third-party electronics, which is a stronger claim than either party could make alone.

Speed is where superconducting hardware separates itself. Current Rigetti systems perform two-qubit gates in 50 to 70 nanoseconds, roughly a thousand times faster than the gate speeds reported for trapped-ion and neutral-atom machines [12]. More operations fit inside a coherence window, which partly offsets the shorter coherence times.

The multi-chip approach is working. Rigetti's Cepheus-1-36Q system is built from four interconnected 9-qubit chiplets and reached 99.5% median two-qubit gate fidelity [13]. That halved the two-qubit gate error rate compared with the earlier 84-qubit single-chip Ankaa-3 system. A smaller assembled machine outperforming a larger monolithic one is the argument for modular construction.

The product line starts well below those systems. The 9-qubit Novera processor, introduced in 2023 on the Ankaa-class architecture with tunable couplers, is the on-premises entry point, and the line runs up through the 108-qubit Cepheus systems [14].

4. IQM

IQM's approach is defined by where its machines go rather than what they achieve. It builds a graded product line and installs it inside national supercomputing centres, which makes deployment count a more meaningful number than peak specification.

Recent Developments

IQM deployed its first U.S. system in June 2026, a 20-qubit Pathfinder Radiance device installed at Oak Ridge National Laboratory for hybrid quantum and high-performance-computing work [15].

Current State

IQM's largest deployed systems are 54-qubit Radiance and Crystal devices [16]. One went to CINECA in Bologna in late 2025, integrated with the Leonardo supercomputer. Another has run at the Leibniz Supercomputing Centre in Munich since February 2026, as Europe's first EuroHPC-funded quantum computer.

The larger tier is announced rather than delivered. As of the Munich deployment, IQM's 150-qubit Radiance tier had not arrived at that site [16]. The 150-qubit upgrade to the 54-qubit system was still planned for the end of 2026.

Gate quality is consistent across the range. The 20-qubit Garnet processor reaches a median two-qubit gate fidelity of 99.5%, with genuine entanglement demonstrated across all 20 qubits at once [17]. The 54-qubit

Emerald system, using the same architecture scaled up, reports the same 99.5% median figure for its two-qubit gates [18]. Holding fidelity constant while more than doubling qubit count is the harder half of scaling.

The product line spans 5-qubit, 20-qubit, 54-qubit and 150-qubit systems, all built on a square-lattice architecture where each qubit connects to its neighbours through a tunable coupler [19].

5. Fujitsu and RIKEN

This collaboration is the clearest demonstration that the limiting factor for superconducting machines is often the refrigerator rather than the chip. Its most recent system quadrupled qubit count inside the same cooling hardware.

Current State

The current-generation system holds 256 qubits and was unveiled in April 2025 at the RIKEN RQC-Fujitsu Collaboration Center [20]. It quadruples the capacity of the 64-qubit machine the pair deployed in October 2023.

The design scaled without a redesign. The 256-qubit system uses a three-dimensional architecture built from repeatable 4-qubit unit cells, the same unit-cell design as its 64-qubit predecessor [20]. Growth came from repeating a proven block rather than drawing a new chip.

The cooling result is the one that generalizes. The system achieves a fourfold increase in implementation density inside the dilution refrigerator, letting four times the qubits operate within the same cooling unit the 64-qubit machine used [20]. Dilution refrigerators are large, expensive and slow to build, so density inside an existing one is worth more than it sounds.

The machine became available to companies and research institutions worldwide during the first quarter of fiscal year 2025, through the pair's hybrid quantum-classical platform [20].

6. Materials and Device Research

The results in this section are not machines. They are attempts to change what superconducting qubits are made of, and they matter on a longer horizon than any product roadmap in the preceding sections.

Recent Developments

MIT researchers developed air-stable ultrathin niobium diselenide superconductors, using encapsulation epitaxy with graphene to keep them stable [21]. The material can be integrated into superconducting microwave circuits at wafer scale. Air stability is the practical barrier with ultrathin superconductors, because a material that degrades on contact with atmosphere cannot go through normal fabrication.

Yale researchers fabricated all-nitride transmon qubits from niobium nitride and aluminium nitride using atomic layer deposition [22]. They achieved microsecond-scale T_1 above 300 millikelvin. The temperature is

the point. Conventional superconducting qubits need to sit well below that, and every millikelvin of headroom reduces the cooling burden described in Section 5.

ETH Zurich coupled a superconducting transmon qubit to a mechanical resonator, specifically a high-overtone bulk-acoustic wave device, and used the hybrid to perform C-PHASE gates and small quantum algorithms [23]. The work was published in Science. Mechanical resonators can hold a state far longer than a transmon, so pairing the two is an attempt to get transmon speed with mechanical memory.

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